

Ant Inspired: Six legs arranged into two groups of three legs to act as a stable tripod. On servo drive each leg. It has a tripod gait where the bottom linkage of the leg is freely moving. When a leg is on the ground pushing, the knee stop locks the shin in place, creating a rigid leg that propels the robot forward. When the leg lifts off the ground, the shin freely folds forward, allowing the leg to swing back compactly through the air without dragging.

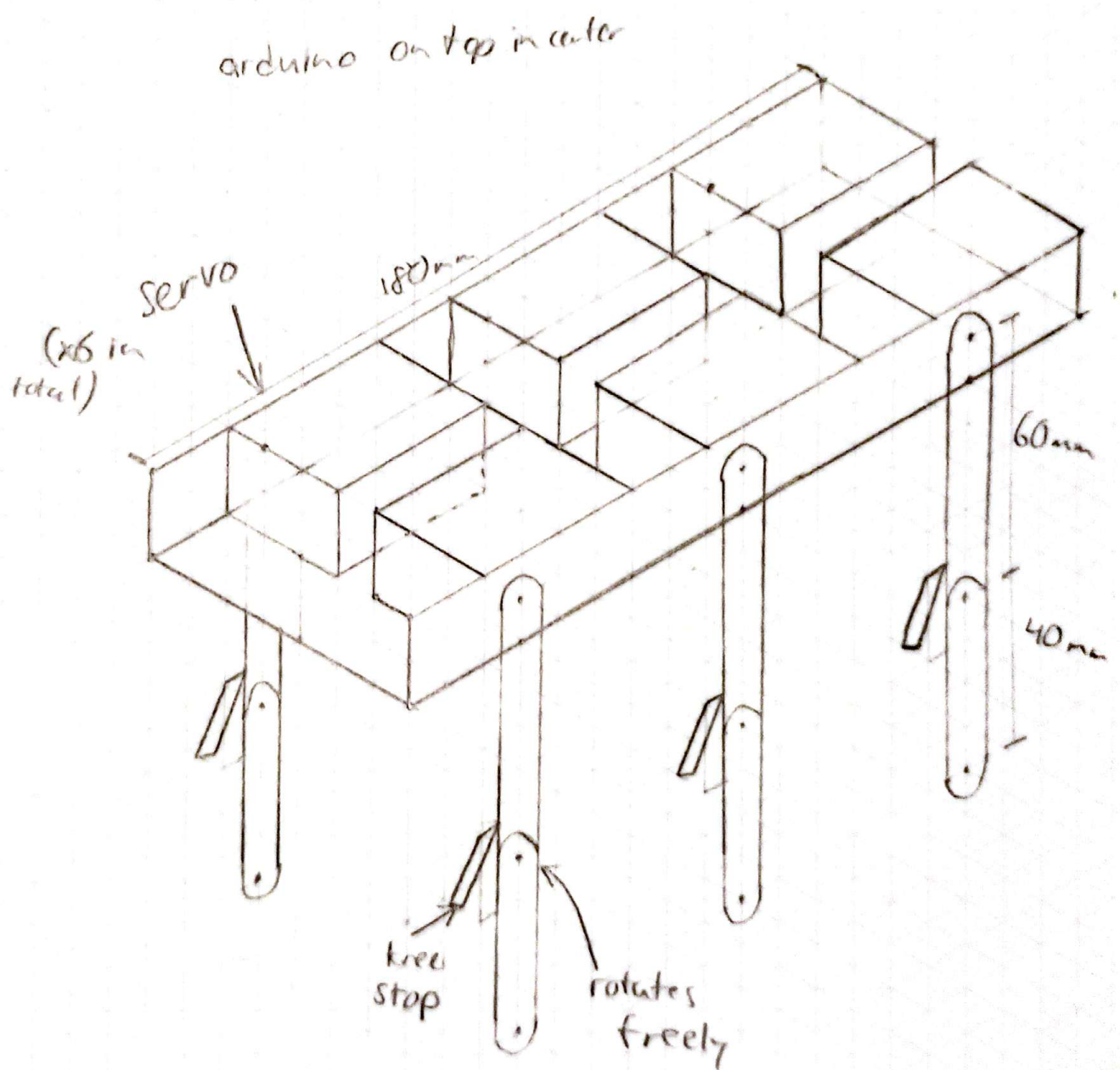
Inchworm Inspired: Two body segments that are connected by an extension mechanism. To push forward, the rear anchor will activate, and the front will deactivate. The servo will rotate causing an extension from a crank-rocker linkage. Then the anchors will switch, and the rear will be pulled forward.

I chose the hexapod (ant-inspired) because it offers the best balance of speed, performance, and reliability. The tripod gait provides static stability as there are three legs always supporting the robot, making it unlikely to tip over. Additionally, the hexapod has a potential for higher speed, so it can compete for bonus points. While it requires six servos compared to the inchworms two, the torque analysis confirms the servos have enough torque to move the weight. The bio inspiration is simple and using the crank-rocker linkages as the legs are likely to be successful. Although it requires more servos, the hexapod is the better choice due to its stability and speed.

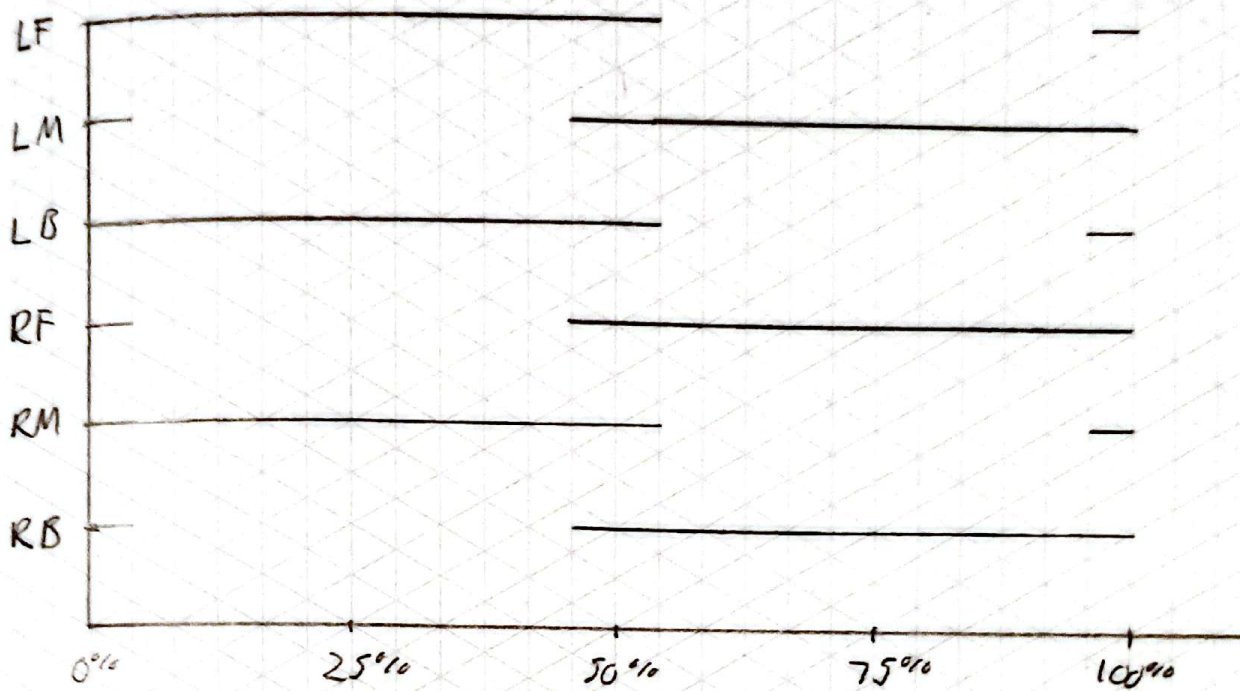
Bio Sketch 1

Ant inspired

→ 1 leg = 10 mm



Bio Gait Plot 1



Stability analysis 1

Component	Mass	X-pos (mm)	Y-pos (mm)
Servos	270g	95	35
MDK	250g	95	35
Arduino	50g	45	35
Battery	100g	135	35
Other	20g	90	35
total	690g		

Center of mass $\rightarrow X: (270 \cdot 95 + 250 \cdot 95 + 50 \cdot 45 + 100 \cdot 135 + 20 \cdot 90) / 690$

$X_{com} = 97 \text{ mm}$ from front

$Y_{com} = 35$ $Z_{com} \rightarrow$ not important

✓ good

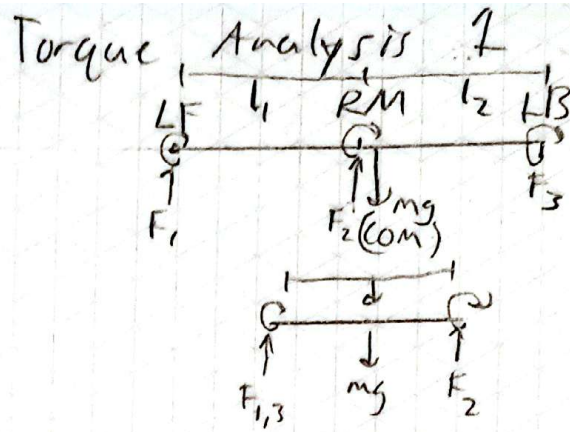
Weight $= 0.690 \text{ kg} \cdot 9.81 \text{ m/s}^2 = 6.77 \text{ N}$



Center of mass inside
of support polygon

$$W = 6.77 \text{ N}$$

$$= mg$$



side view

Front view

$$\Sigma F = 0$$

$$F_1 + F_2 + F_3 = mg$$

$$r = 0.06 \text{ m}$$

$$\Sigma M = 0$$

$$d(F_2) = \frac{1}{2} d mg \quad F_2 = \frac{1}{2} mg \rightarrow F_1 + F_3 = \frac{1}{2} mg$$

$$l_1 = l_2 \rightarrow F_3 = \frac{mg}{4} = F_1$$

$$F_2 = 3.39 \text{ N}$$

$$T = F \cdot r \quad r = 0.06 \text{ m}$$

$$T = 3.39 \text{ N} (0.06 \text{ m}) = 0.2 \text{ Nm}$$

$$\frac{0.2}{0.0981} = 2.07 \text{ kg} \cdot \text{cm}$$

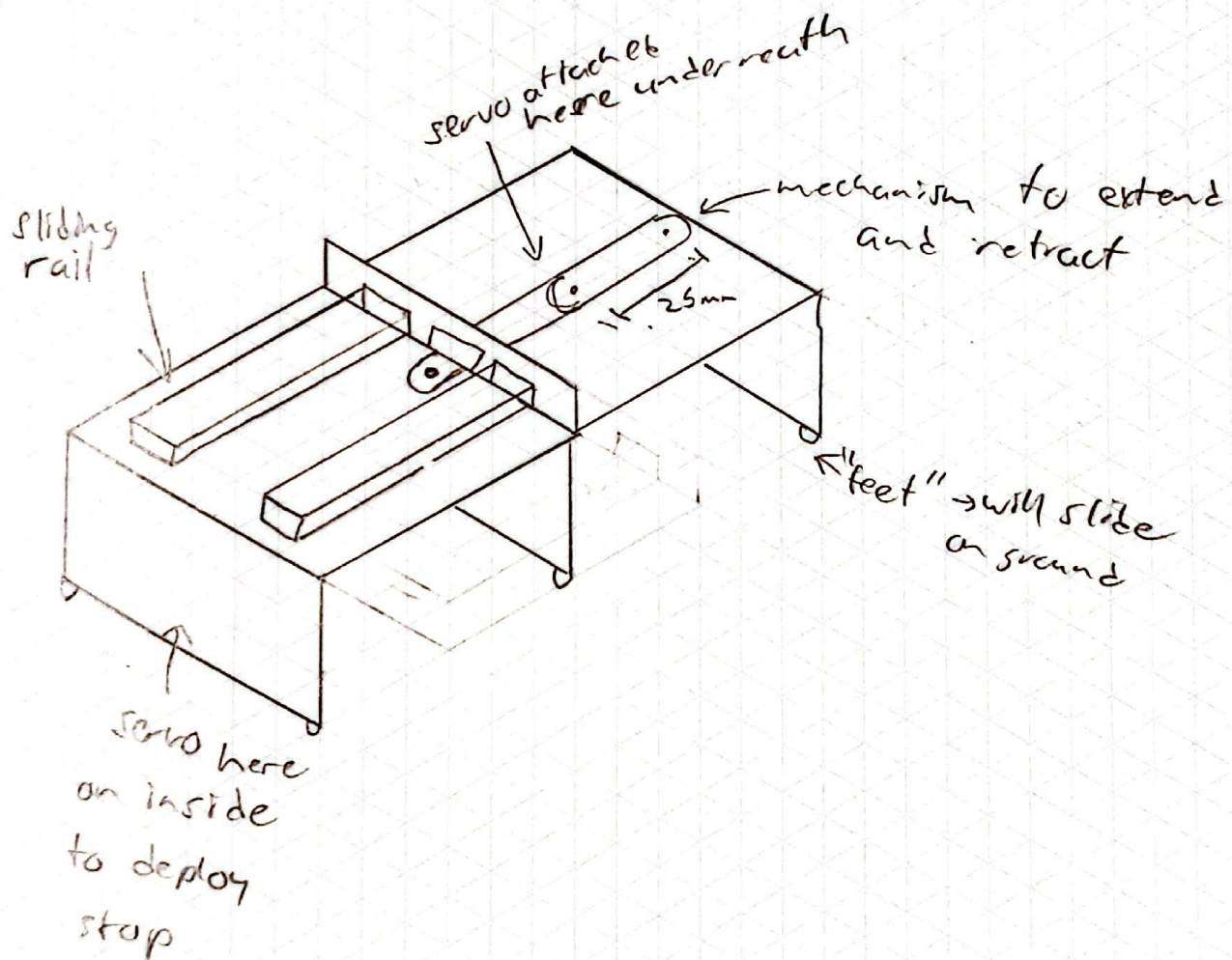
$$2.07 \text{ kg} \cdot \text{cm} = T_{\text{max}}$$

Servo can do $4.1 \text{ kg} \cdot \text{cm}$ so ✓

Bio Sketch 2

Inchworm inspired

May do cutouts to
reduce weight



Gait plot 2



stability analysis 2

Component	mass
2 servos	135g
MBF	200g
Arduino	80g
Battery	100g

$$\text{Total mass} = 485g$$

$$\text{Weight} = 485 \cdot 9.81 = 4.75N$$

The mechanism is connected to ground always in the 4 corners, so it does not matter where the COM is, it is always inside of contact points

Torque analysis 2 side

$$W = 4.75N$$

$$\sum F = 0$$

$$F_1 + F_2 = mg$$

$$F_1 = F_2 = \frac{1}{2} mg = 2.375N$$

$$T = F \cdot r$$

$$r = 0.03m$$

$$T_{1,2} = 2.375 \cdot 0.03 = 0.0713 Nm = 7.3 kg \cdot cm \checkmark$$

$$T_3 = T_1 = T_2$$

$$T_3 = \frac{1}{2} mg \cdot r = 7.3 kg \cdot cm \checkmark$$

